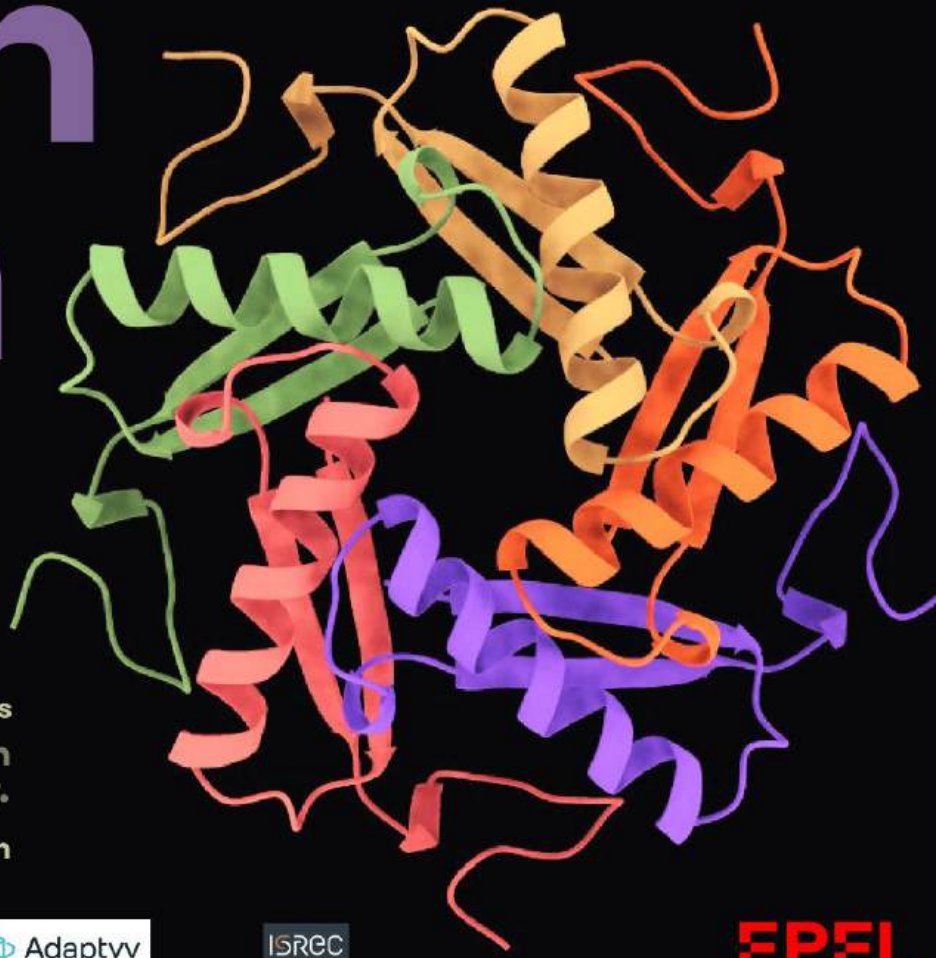
designinglifewithai.ch

Protein design week.



Weekend Hackathon

From **AI Designs** to **Lab Prototypes** in 48 hours.

DLAI – Association Reconnue par l'EPFL

March 2–8, 2026 | EPFL Campus

7 Days of **AI-Driven Protein Engineering.**

designinglifewithai.ch

Cradle · lemnabio · Hugging Face · Adaptyv · ISREC · EPFL

About Us

Designing Life with AI

EPFL MAKE: Student-led research project

Design, model, and experimentally test novel proteins using computational tools and wet-lab methods

SV Industry

Organize conferences and events to bridge the gap between SV students and industry

Commission of the Association of Life Sciences students (AESV)

Protein Design Week 2026

- Goal : promote AI protein design at EPFL
- Foster a **fun learning** environment with people from diverse backgrounds
- During the week : workshop, office hours and lab visit
- This weekend : **36h hackathon**

Program

- Monday: **Introduction & Workshop** ✨
- Tuesday: **Office Hours** 💻
- Thursday: **Adaptyv Bio Lab visit** 🧪
- Friday, 18:30: **AI-driven Protein Design: Innovation Unleashed or Biosafety Nightmare?** 🧬
- Saturday & Sunday: **Hackathon** 💥

PDW

Resources

- Centralized communication on **Slack**: announcements, team channels, etc...
  Please join the Slack now !!!
- Centralized computing resources on **Github:** explanations, code snippets

→ **still not complete**

Slack

Github

PDW

TO DO list

Before the hackathon...

- ✅ **Install PyMol** (or other protein visualization tool): https://pymol.org/
- ✅ **Install Docker, Kubectl and Run:ai** (EPFL students only): tutorial here (Part 1)
- ✅ **Create a HuggingFace account:** https://huggingface.co,
  Submit your Hugging Face username here for expanded rate limits
  Join the Hugging Science organization: HuggingScience
- ✅ Fill this form so we can use Alphafold3 without getting sued by Google Deepmind. (**mandatory**)
- ✅ Read your associated **survival guide on Github.** (optional)
  Come to **office hours** tomorrow if you have any questions (very optional).

PDW

Let’s start with some biology…

Do you know what’s a protein?

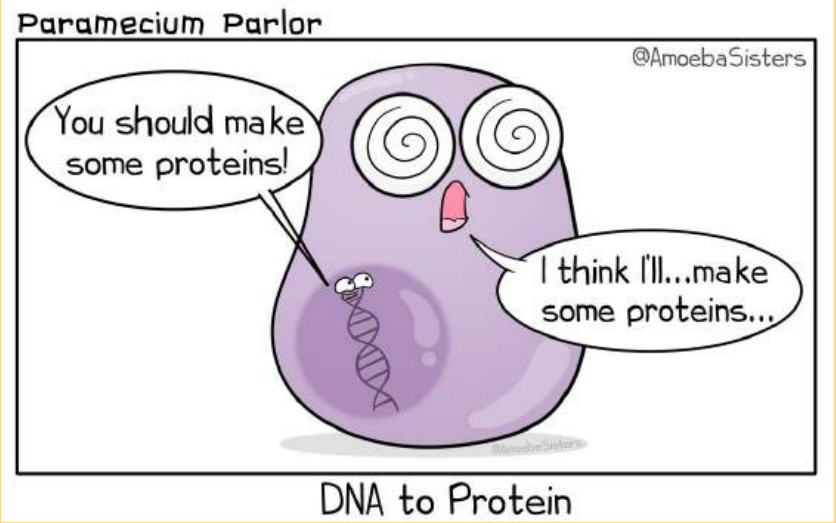
PDW

What is a protein ?

Basics

- Proteins are **functional macromolecules** composed of **amino acids**.
- Their sequences are **encoded in our DNA**.

DNA is like a cookbook;
Genes are the recipes,
Proteins are the food.



https://www.youtube.com/watch?v=oefAI2x2CQM

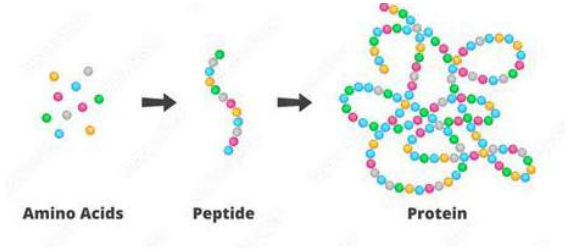
PDW

What is a protein ?

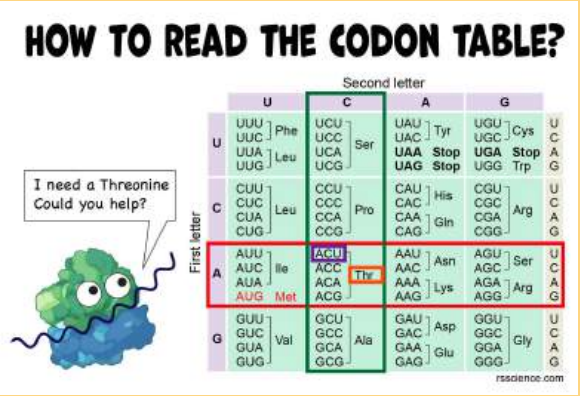
- DNA codes for amino acids through **codons**:

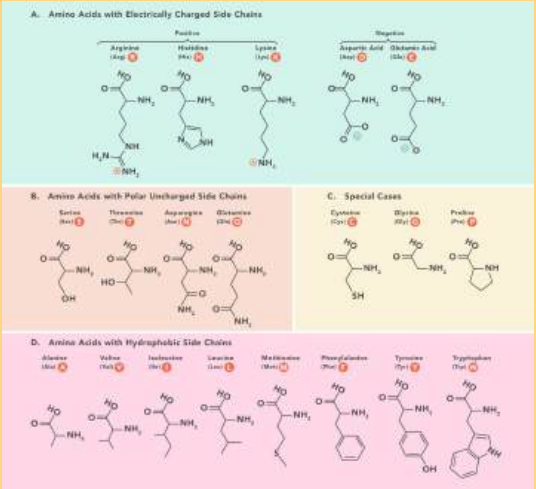
  A, T, C, G → 20 A. A.

- Proteins synthesized as **polymers of amino acids**, like beads on a string, that spontaneously fold into specific shapes:



DNA → RNA → Protein → …



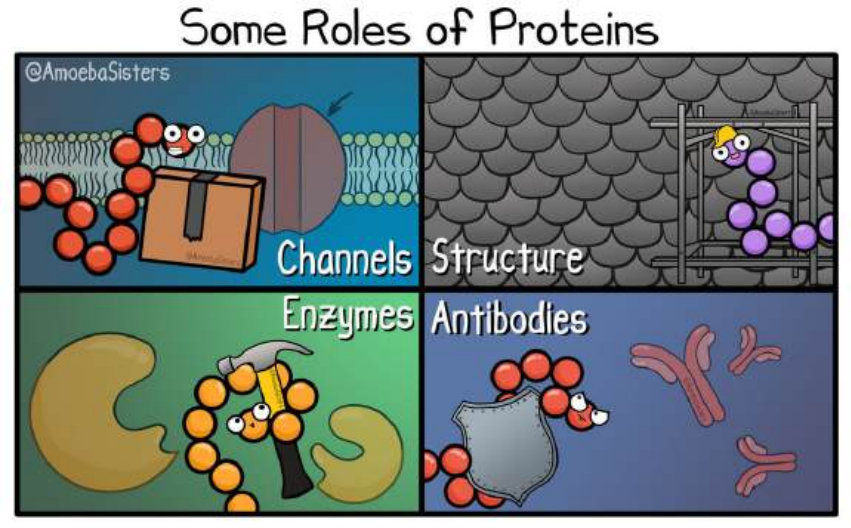


What is a protein ?

Functions

Proteins do nearly everything in cells:

- **Enzymes** speed up reactions, transporters move molecules, and receptors sense signals.
- **Structural proteins** like collagen give shape and strength to tissues.
- Even movement, like muscle contraction or cell division, relies on protein machinery.



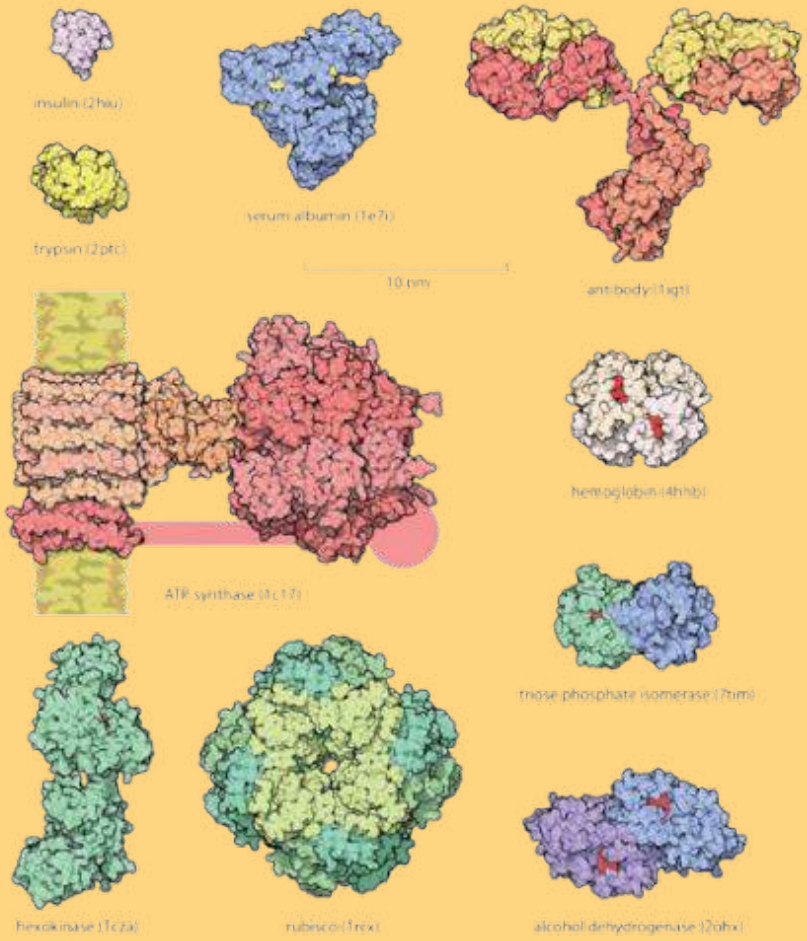
PDW

What is a protein ?

A protein's 3D shape directly dictates what it does

- The folding pattern brings the right atoms close for specific chemical interactions.
- Mutations may change shape and disable, alter, or improve function.
- Protein adopting wrong conformations (misfolding) cause diseases.

Structure

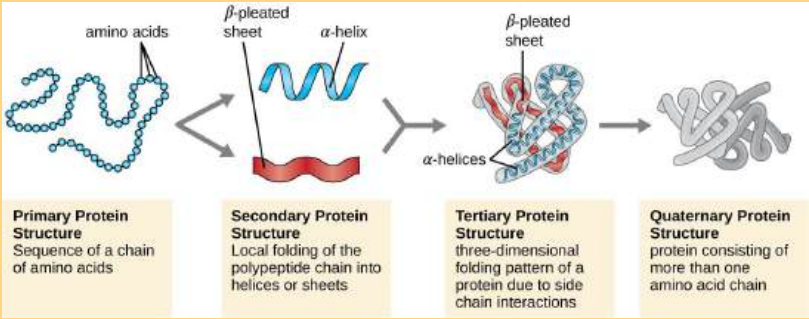


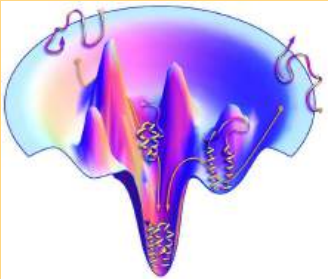
PDW

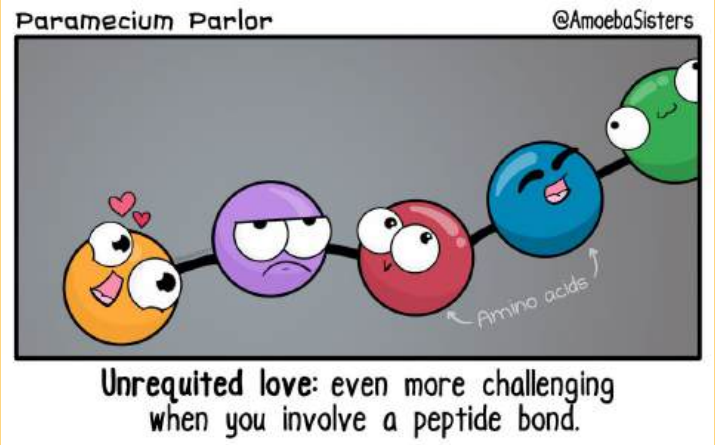
The protein folding problem

Proteins are expressed as amino acid chains, then folded and further modified into complex structures.

- The folding occurring minimizes the energy of the protein (energy landscape idea).
- Each amino acid can move in countless ways, creating astronomical folding possibilities.
- The environment (water, pH, ions, crowding) strongly influences the final shape.
- **Predicting structure means solving physics at the atomic level; an extraordinary computational task.**



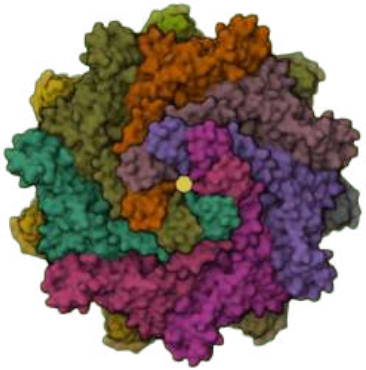




PDW

AI as a solution ?

AEPVYPDQLRLFSLGQGVCGDKY
RPVNREEAQSVKSNIVGMMGQW
QISGLANG…



PDW

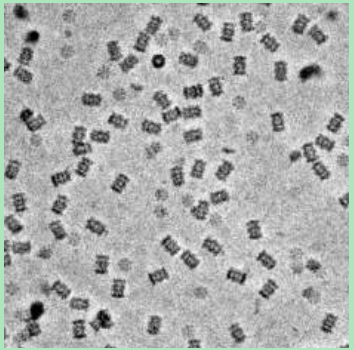
AI as a solution ?

- AI / ML models need training on a **large database**
- In the case of protein design, they use the **Protein Database (PDB)**
- **Structures are characterized experimentally**: X-ray crystallography, NMR, (cryo) electron microscopy, …

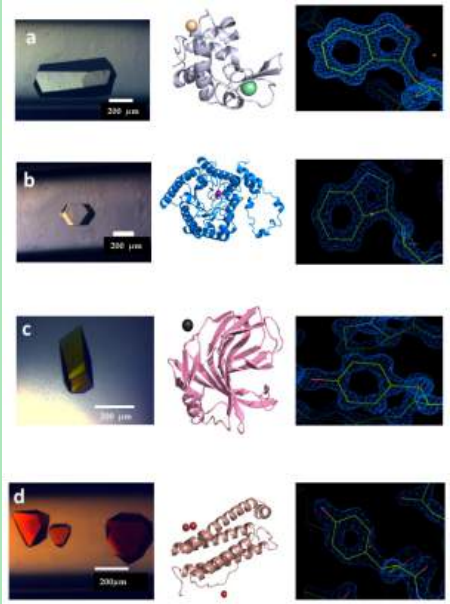
Data base

RCSB PDB PROTEIN DATA BANK

249,906 Structures from the PDB archive



Electron microscopy



X-ray crystallography

PDW

AlphaFold

AI system that predicts the 3D structure of a protein from its amino acid sequence.

- It uses deep learning to learn patterns from many known protein structures and sequences.
- In many cases, its predictions are about **as accurate as experimental structures**.
- AlphaFold has made it possible to **compute structures for millions of proteins that have never been experimentally solved.**

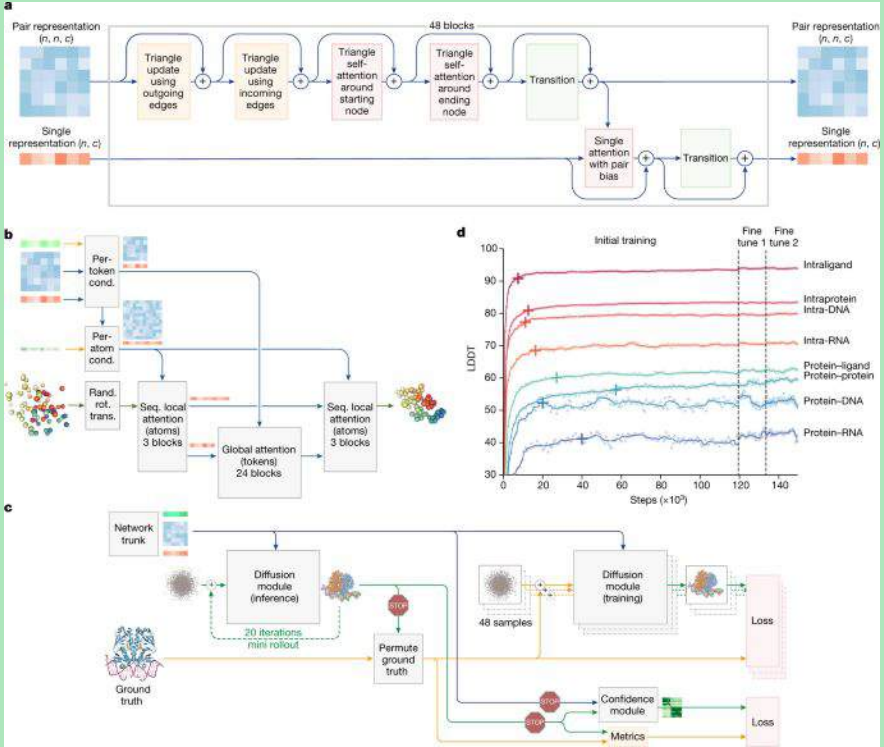
Abramson, J., Adler, J., Dunger, J. *et al.* Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024). https://doi.org/10.1038/s41586-024-07487-w

In theory …

RCSB PDB PROTEIN DATA BANK

249,906 Structures from the PDB archive

1,068,577 Computed Structure Models



PDW

AlphaFold

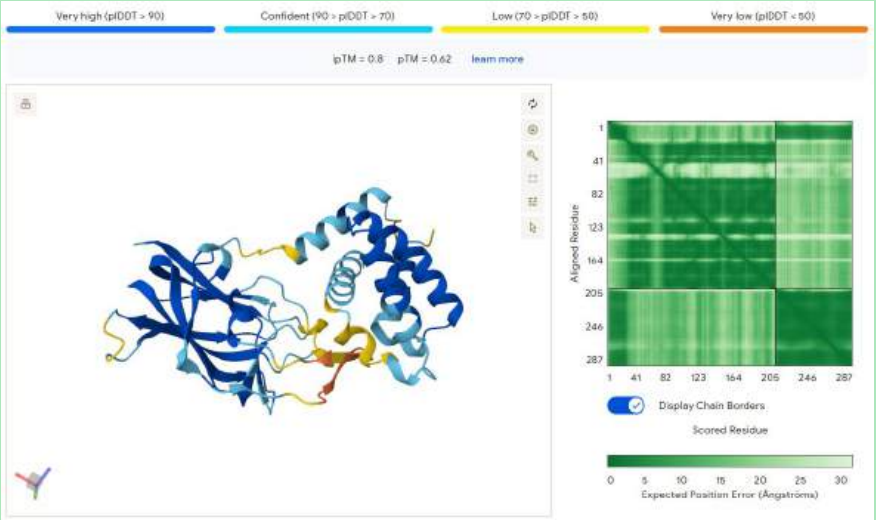
Online server

- Online server: easy to use
  - 30 runs per day
  - Need a google email address
    (! doesn't work when connected with epfl email)
- Locally: needs a large downloaded database
- Other models: Rosetta Fold, Boltz, …

Abramson, J., Adler, J., Dunger, J. *et al.* Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024). https://doi.org/10.1038/s41586-024-07487-w

PDW

In practice …



- **Confidence matrix:** grid showing how sure the model is about relative positions of all pairs of residues/atoms.
- **plDDT**: per-atom/local confidence score (0–100) for how reliable the local structure is.
- **pTM**: global confidence for overall fold/topology of the whole structure.
- **ipTM**: confidence that interfaces/interactions between different chains or partners are correct.

https://alphafoldserver.com/

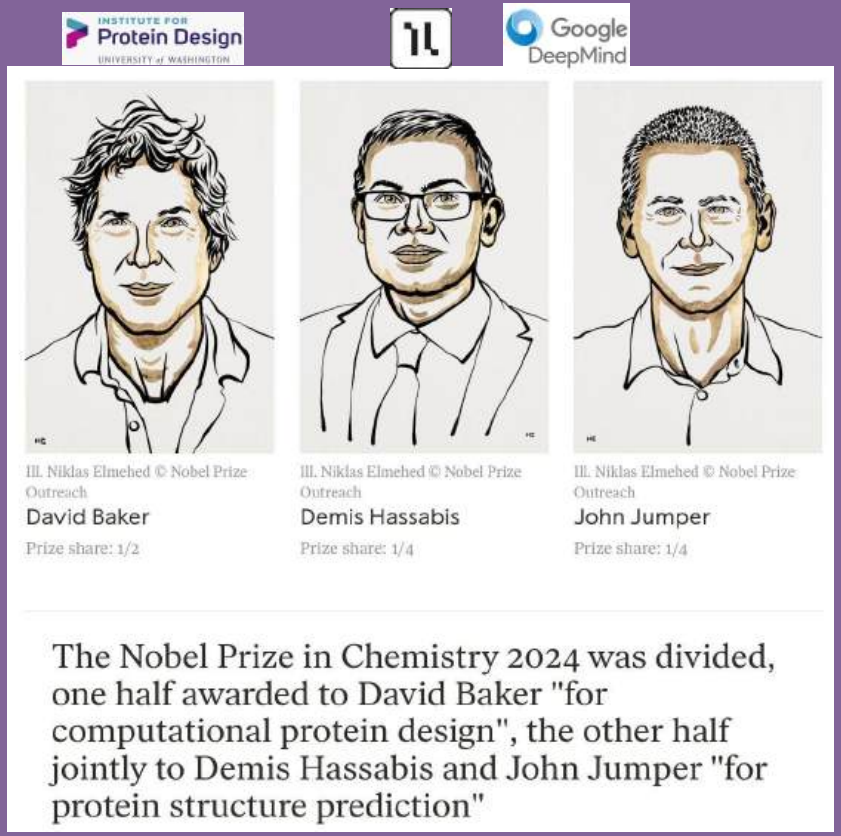
Why is that cool?

AI-based approaches can **predict** :

- The **structure** of a folded protein from its amino acid sequence in **seconds**.
- The **interactions** with other proteins, nucleic acids and small molecules.

Allowing to **generate** **novel or modified structures and sequences** to achieve given functions.

Nobel Prize in Chemistry 2024



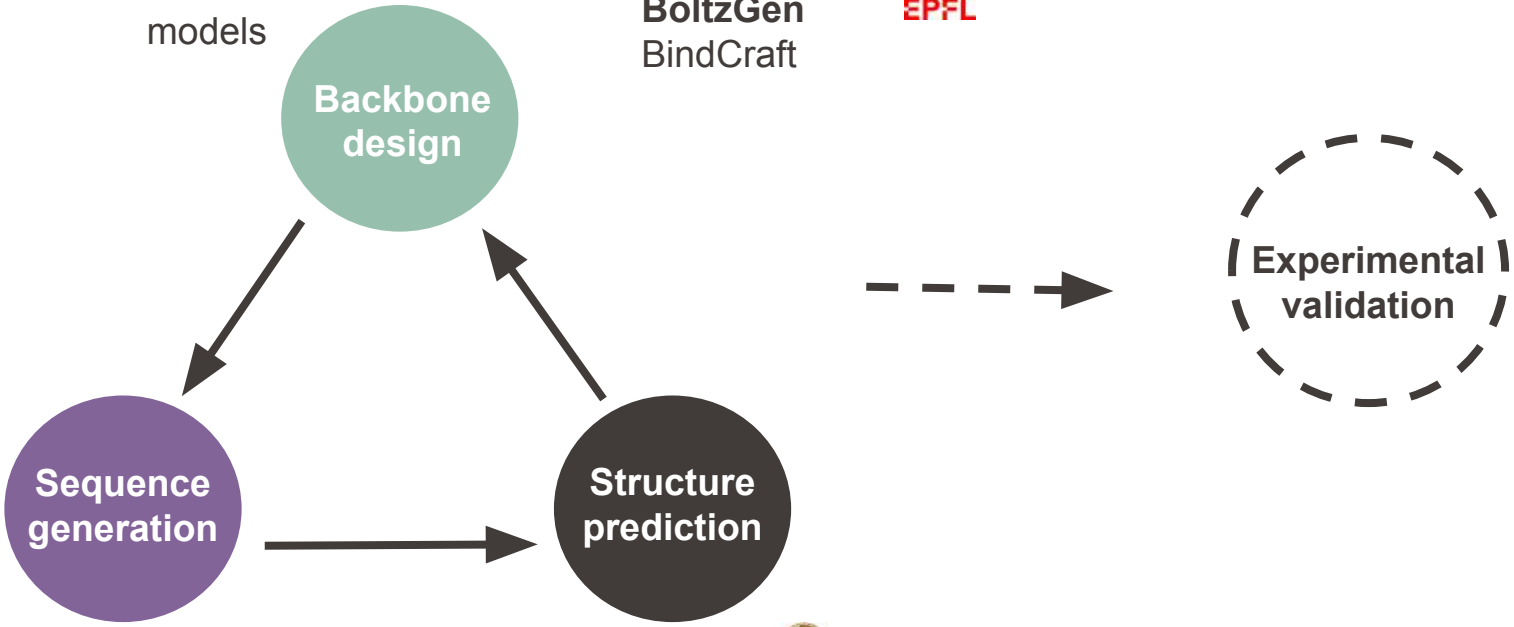
PDW

Protein design

From predictive to generative models - Typical pipeline overview

RFdiffusion
Other diffusion or hallucination models

Full pipeline models:
BoltzGen EPFL
BindCraft



LigandMPNN (ProteinMPNN)
CARBonAra EPFL

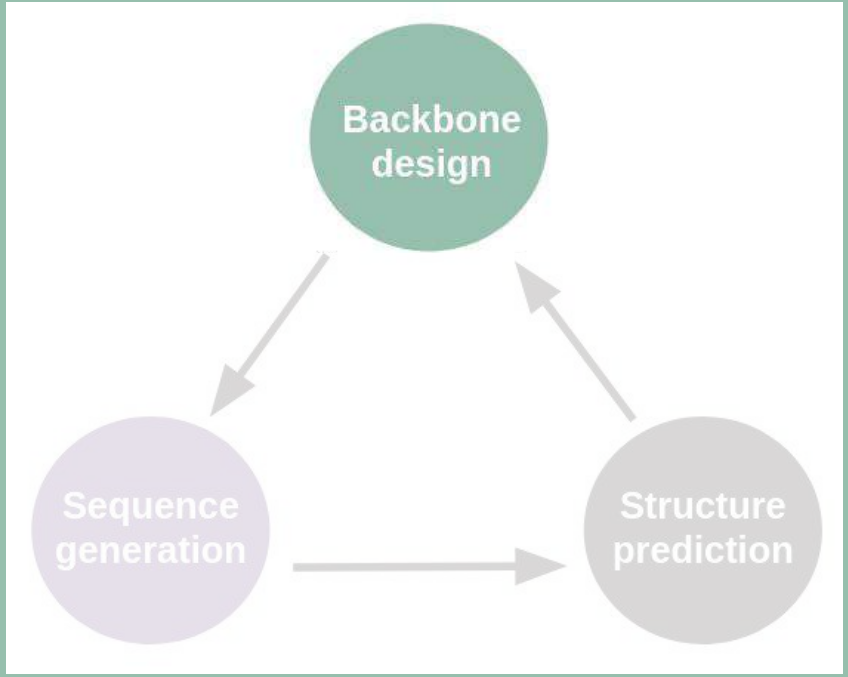
AlphaFold

PDW

Step 1: Backbone design

Protein design

- Defines the **3D scaffold** of a protein before an amino acid sequence is assigned.
- Focuses on specifying secondary structure elements and their spatial arrangement.
- Requires sampling conformations consistent with physical and geometric constraints (bond lengths, angles, sterics).



PDW

RFDiffusion (3)

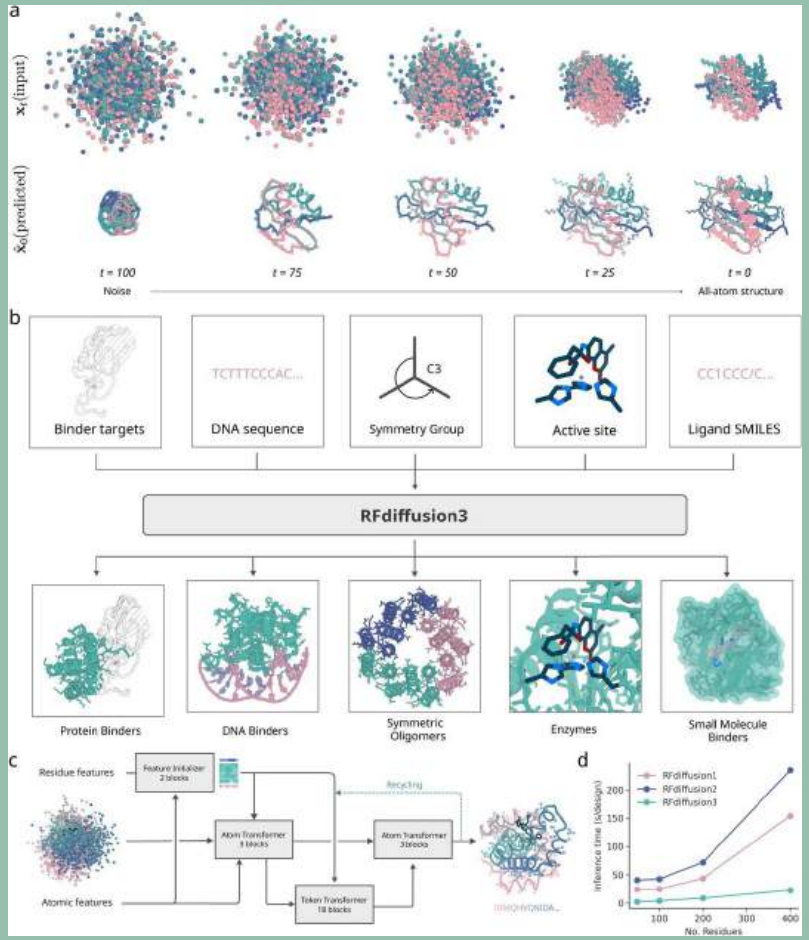
Implemented for PDW Hackathon

- RFdiffusion uses a **diffusion model** to generate **de-novo** protein backbones.
- Iteratively "denoises" random noise into structured protein geometries guided by learned protein priors.
- Allows **conditional generation**, e.g. scaffolding around binding sites or motifs.

Watson, J.L., Juergens, D., Bennett, N.R. et al. De novo design of protein structure and function with RFdiffusion. Nature 620, 1089–1100 (2023). https://doi.org/10.1038/s41586-023-06415-8

Butcher et al. De novo Design of All-atom Biomolecular Interactions with RFdiffusion3. Preprint at https://doi.org/10.1101/2025.09.18.676967 (2025)

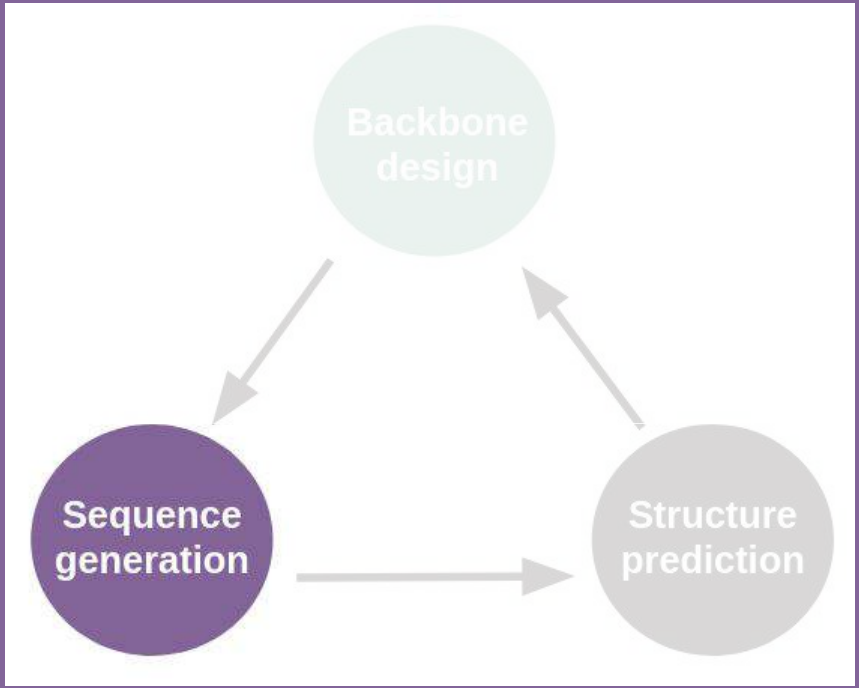
Protein design - Backbone design



Step 2: Sequence generation

- Assigns **amino acids** to the designed backbone to ensure structural stability and function.
- Balances **foldability**, **energy minimization**, and **functional constraints**.
- One protein structure can have **multiple possible sequences** that may have **different properties**.

Protein design



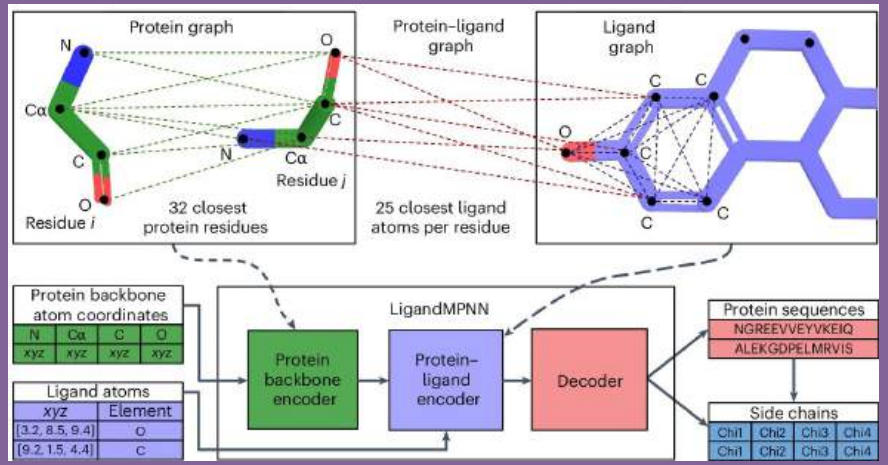
PDW

LigandMPNN

Implemented for PDW Hackathon

Protein design - Sequence generation

- Extension of ProteinMPNN optimized for protein–ligand interface design (multiple “modes”).
- Uses message passing neural networks over backbone graphs to assign amino acids.
- Learns **geometric and energetic patterns** that yield stable and functional binding pockets.



Dauparas, J., Lee, G.R., Pecoraro, R. et al. Atomic context-conditioned protein sequence design using LigandMPNN. Nat Methods 22, 717–723 (2025). https://doi.org/10.1038/s41592-025-02626-1

PDW

CARBonAra

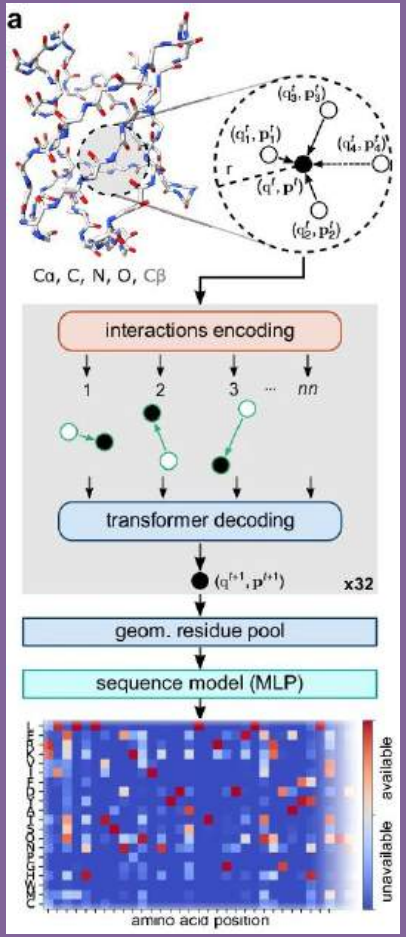
Implemented for PDW Hackathon

- Uses a **geometric transformer** architecture to predict amino acid probabilities.
- Direct prediction from backbone atom **coordinates** and any **surrounding molecular context**, including ligands, ions, nucleic acids, or other proteins.
- Ideal for functional designs like enzyme engineering under molecular constraints.

Krapp, L.F., Meireles, F.A., Abriata, L.A. et al. Context-aware geometric deep learning for protein sequence design. Nat Commun 15, 6273 (2024). https://doi.org/10.1038/s41467-024-50571-y

PDW

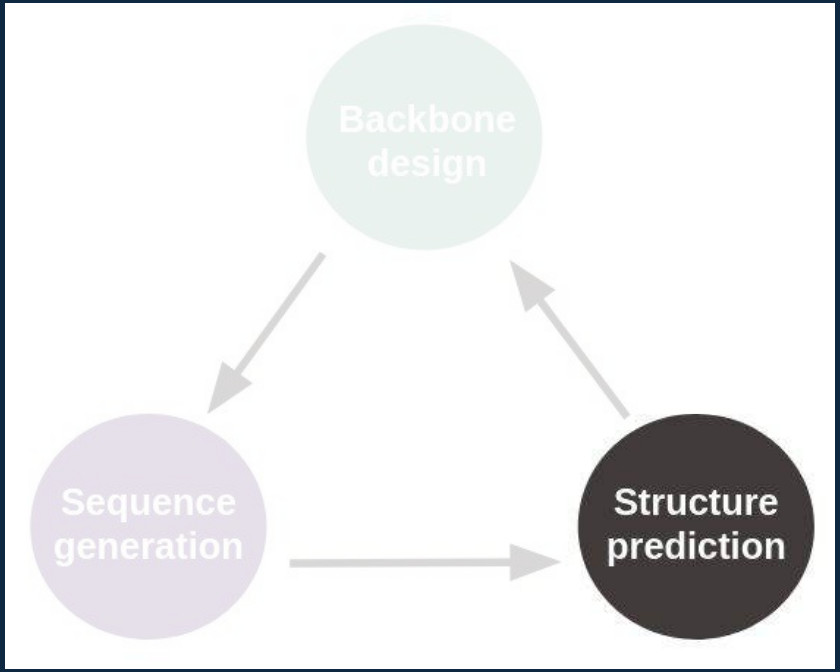
Protein design - Sequence generation



Step 3: Folding validation

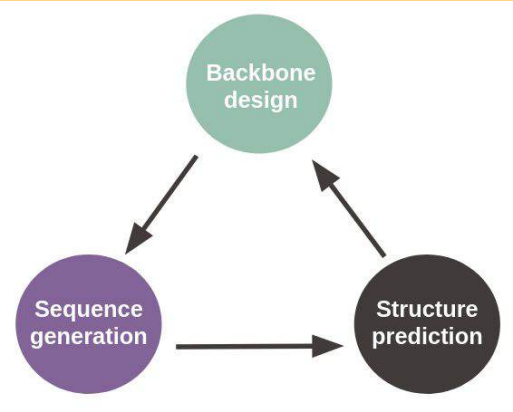
- Protein structure prediction is performed to check that the generated backbone folds into the wanted conformation.

- *See AlphaFold and alternatives*



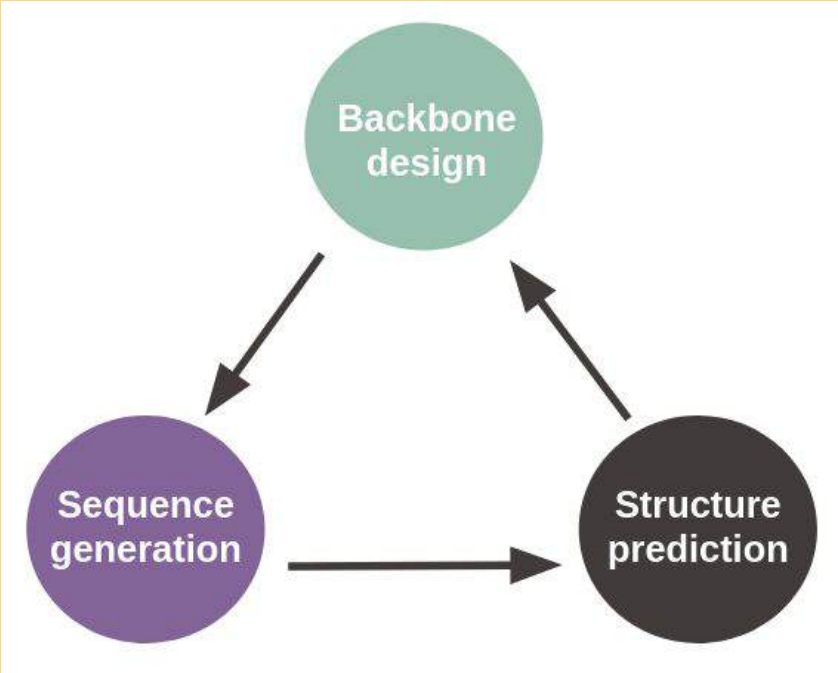
Combining the steps

- To perform a full protein design task, you will need to “assemble” the models into a **pipeline**.
- E.g. RFDiffusion is often coupled with ProteinMPNN and RosettaFold (all from the same lab).
- In practice, many different configurations can be tested.



Full pipeline models

- Some published models combine directly the 3 steps; one command runs everything.
- Some filtering or result evaluation is often performed directly in the model.
- These models are more **computationally expensive** (longer runs). The limiting factor is structure prediction.



BindCraft

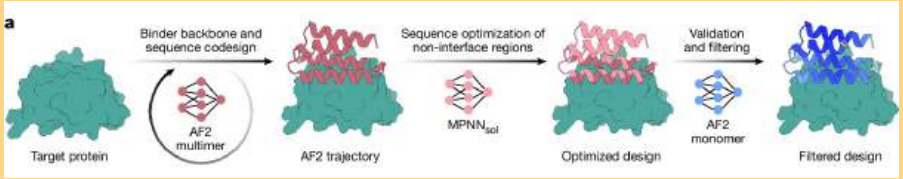
1. Backbone generation: hallucination
2. Sequence prediction: ProteinMPNN
3. Folding: AlphaFold 2

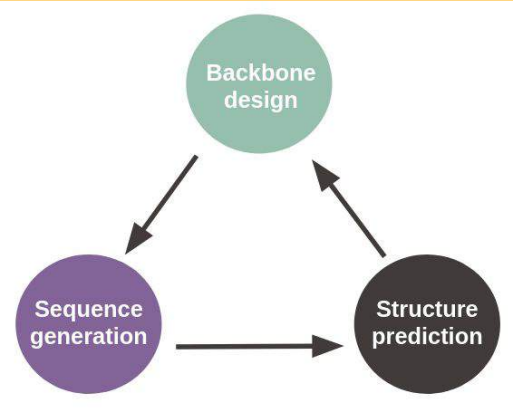
- Includes **strong filters**, resulting in predictions with **high success rates**.

- Is very computationally expensive.

Pacesa, M., Nickel, L., Schellhaas, C. et al. One-shot design of functional protein binders with BindCraft. Nature 646, 483–492 (2025). https://doi.org/10.1038/s41586-025-09429-6

Protein design - Full pipeline models





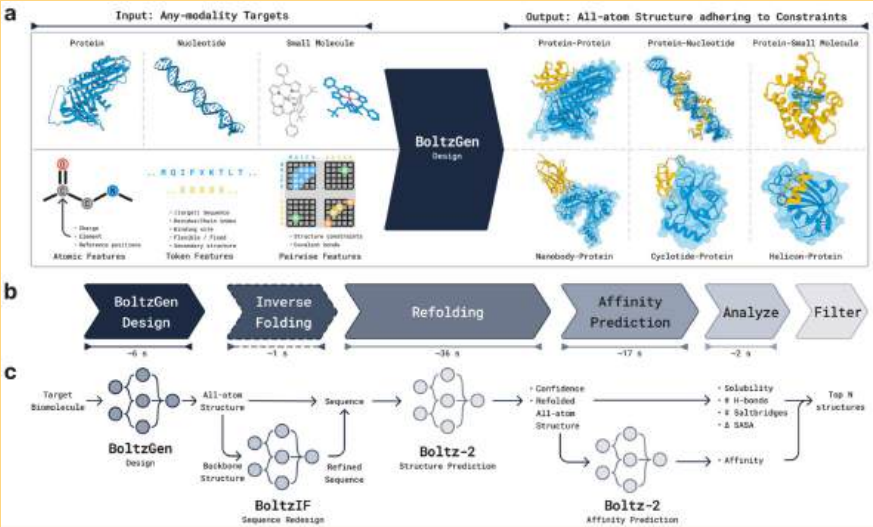
BoltzGen

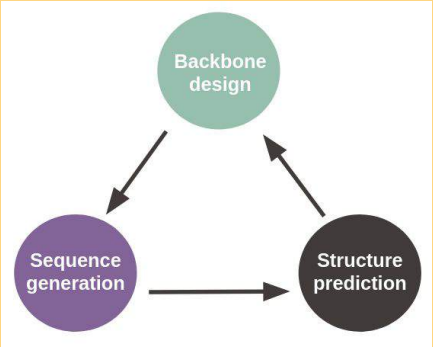
Implemented for PDW Hackathon

- Backbone, sequence and structure all performed with Boltz specific models.
- Includes **multiple design modes** depending on the design task.
- Includes an **affinity prediction module** for small molecule ligands.
- Not as strict filters, not as computationally expensive.

Stark, H. et al. BoltzGen: Toward Universal Binder Design. Preprint at https://doi.org/10.1101/2025.11.20.689494 (2025).

Protein design - Full pipeline models





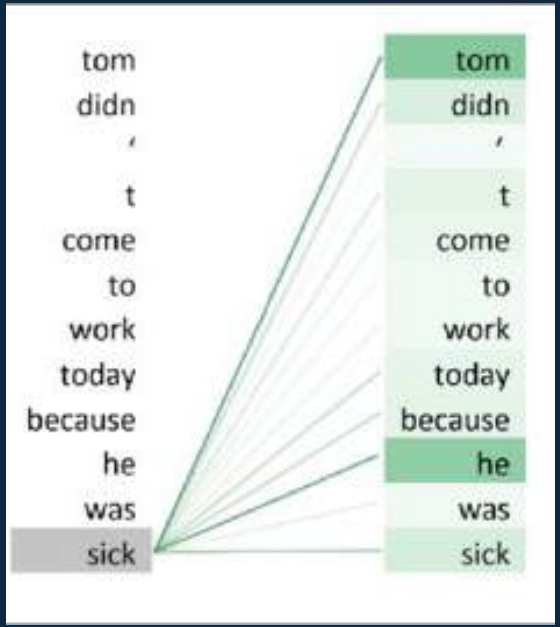
Language Models

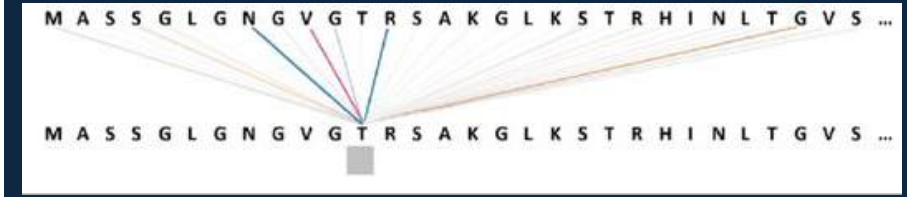
- Language models reason in sequence space, they are based on the **transformer architecture** and the idea of **attention**. it powers today's ChatGPT revolution.

- In the same way, ChatGPT learns grammatical patterns and meaning, protein language models **treat amino acid sequences like words** and learn structural patterns, motifs and protein functions **without structural information**.

- They can **generate new sequences without explicitly building a backbone**, rank candidates, help visualize relationships between proteins...

Alternative design pipeline





PDW

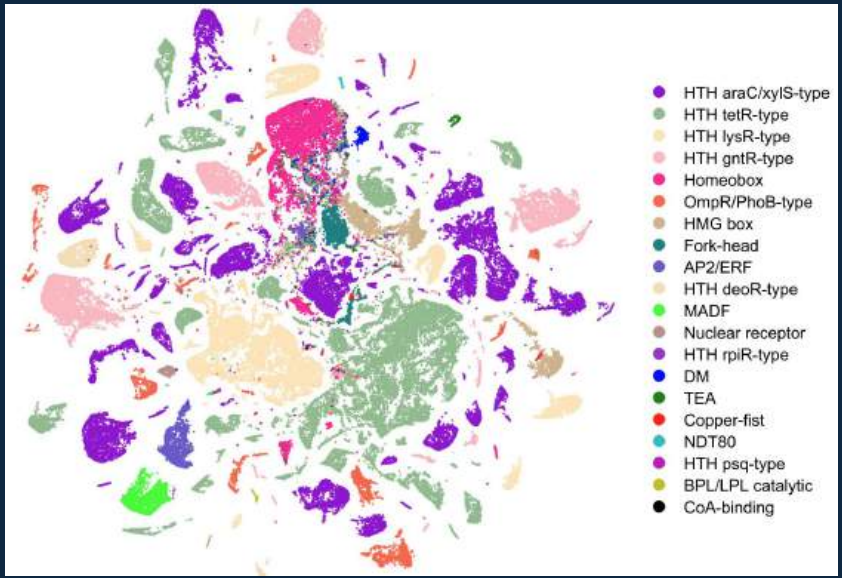
ESM-2

ESM-2 is a **transformer**-based protein language model trained on 250 million+ protein sequences.
Unsupervised training via masked language modeling.

Usage:

- Give it a protein and ask it how natural it is based on its learned distribution → **pseudo-perplexity score**
- Give it a protein and ask for its numerical representation for visualization or downstream prediction → **embeddings**
- prompt it to create a new protein with some conditions → **partial masking**, not just “make me a protein”

Zeming Lin et al. ,Evolutionary-scale prediction of atomic-level protein structure with a language model.Science 379,1123-1130(2023).DOI:10.1126/science.ade2574



t-SNE visualization of DNA-binding proteins using fine-tuned ESM.

Zeng, W., Dou, Y., Pan, L. et al. Improving prediction performance of general protein language model by domain-adaptive pretraining on DNA-binding protein. Nat Commun 15, 7838 (2024). https://doi.org/10.1038/s41467-024-52293-7

PDW

From design back to function …

Filtering & Function analysis

PDW

Filtering designs

How do we know our design is good?
No one-size fits all solution

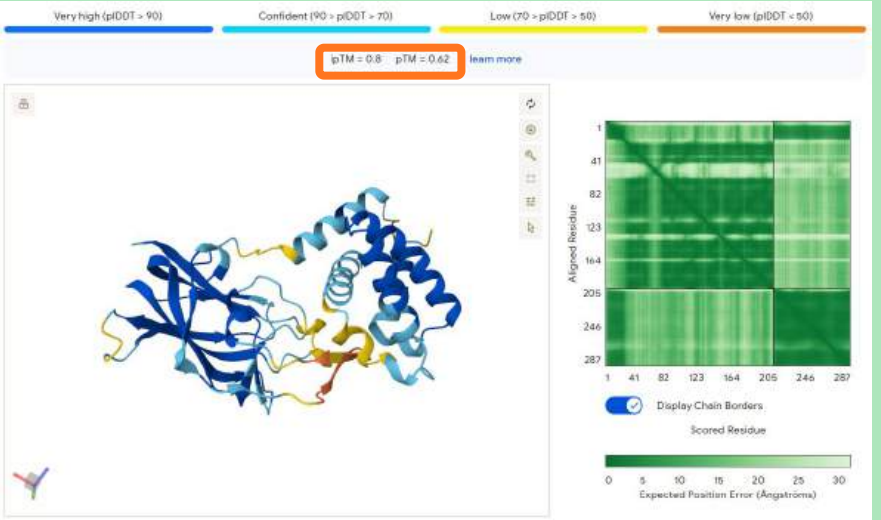
AF3 scores:
- pTM: folding confidence
  high score means well folded

- ipTM: interaction score
  high score means high interaction probability

ipSAE score: Improved metric, similar use as ipTM

Limitation: Scores are based on predicted structures. Sometimes, AF3 predicts **physically impossible folds** → **check for clashes** (residues closer than ~1 Å)

Fold-based filtering criteria



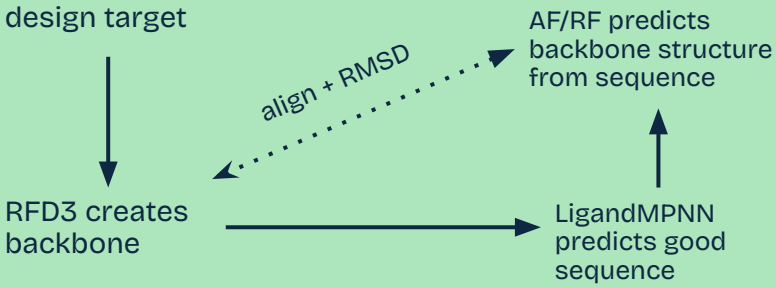
Filtering designs

Choose a subset of generated sequences to express in the lab → need in-silico filtering criteria.

- **Folding validation**, align initial RFDiffusion-generated backbone and AlphaFold prediction and check whether they align (criteria on RMSD)
- **Pseudo-perplexity scores** from protein language models like ESM
- **Specialized models** to predict fitness, stability, toxicity, ...

Other filtering criteria

red: your RFDiffusion3 backbones
yellow: prediction of AlphaFold on your LigandMPNN designed sequence



PDW

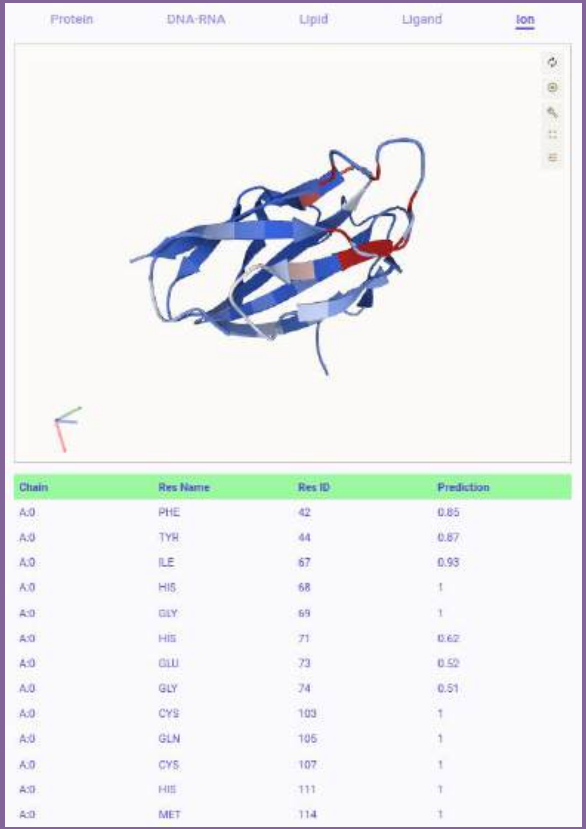
Pesto

Online server

Predict protein function

PeSTo (Protein Structure Transformer) is a geometric deep learning model that **predicts protein binding interfaces** directly f**rom atomic coordinates** (PDB file).

- It allows parameter-free prediction of protein **interactions with proteins, nucleic acids, lipids, ligands and ions**.
- Handles dynamic structures (e.g. MD ensembles) efficiently due to low computational cost, revealing transient interfaces hidden in static crystal structures.
- Serves as the geometric backbone for CARBonAra, adapting interface prediction to context-aware amino acid recovery.



Chain	Res Name	Res ID	Prediction
A:0	PHE	42	0.85
A:0	TYR	44	0.87
A:0	ILE	67	0.93
A:0	HIS	68	1
A:0	GLY	69	1
A:0	HIS	71	0.62
A:0	GLU	73	0.52
A:0	GLY	74	0.51
A:0	CYS	103	1
A:0	GLN	105	1
A:0	CYS	107	1
A:0	HIS	111	1
A:0	MET	114	1

https://pesto.epfl.ch/

PDW

Let’s talk about the hackathon!

PDW

Teams and Projects

- **12 teams of five people**
- Each will work on a different **project proposed by a partner lab or company**
- **Pitch presentations on sunday:** convince the jury your project is the best!

	SATURDAY	SUNDAY
08:30	Doors Open Breakfast, registration	Doors open, breakfast
09:00	Welcome Talk Projects reveal	Hack session 4
09:30	RCP tutorial	
10:00	Hack session 1	
10:30		
11:00		
11:30		
12:00		
12:30		
13:00	Lunch	Lunch
13:30		
14:00	Hack Session 2	Hack session 5
14:30		
15:00		Pitch presentations
15:30		
16:00		
16:30		Jury deliberation
17:00		Prizes
17:30		Apéro
18:00		
18:30		
19:00	Diner	
19:30		
20:00	Hack session 3	
20:30		
21:00		

PDW

Coaches

- PhDs in EPFL's computational protein design labs, members of DLAI, external PhDs.

- Here to discuss your projects with you and orient your design strategies, can help you improve your job configurations, propose next steps.
- Will not help you debug routine python code
- LLMs are pretty good at Pymol, RFD configurations, etc. Ask them first.

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19:30		
20:00		
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21:00		

PDW

Tech stack

- these models are too big to run on your own laptop → will become protein design month...
- **no-code / low-code option:** Hugginface Spaces
- **advanced option:** EPFL's Research Computing Platform

Hugging Face

Research Computing Platform

what are these infrastructures so powerful for AI? GPUs + parallelization

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Hugging Face Spaces

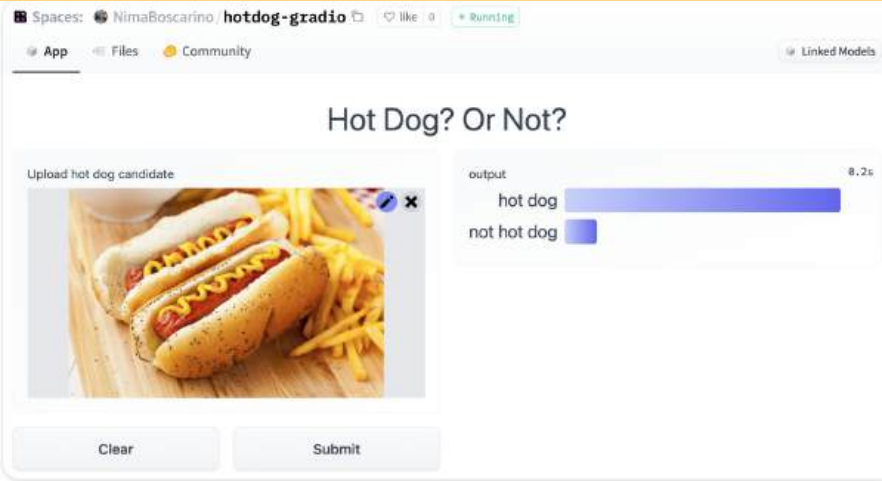
easy way to deploy ML demos

Advantages:

- no setting up computing infrastructure and dealing with dependencies
- easy to use interface (or is it?)

Disadvantages:

- limited flexibility with inputs
- no quick way to integrate into pipelines without manual handling
- not suitable for big jobs (rate limits)
- if you close or refresh the tab the job is gone (but there's a fix!)



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Hugging Face Spaces

Example: RFDiffusion3

Example files for these three tutorials can be found in the `examples` directory of this repo. They are all taken from RFD3's amazing Github repository. This space pairs well with the companion LigandMPNN space for inverse folding!

Config file: .yaml or .json

Drop File Here
- or -
Click to Upload

Target/Scaffold PDB

Drop File Here
- or -
Click to Upload

Number of Designs per Batch

2

Number of Batches

5

Advanced Options

Run Generation

tutorial on RFD3 config coming soon!

```
{
    "M0255_1mg5_unfixed": {
        "input": "./M0255_1mg5.pdb",
        "ligand": "NAI,ACT",
        "unindex": "A108,A139,A152,A156",
        "length": "180-200",
        "select_fixed_atoms": {
            "A108": "ND2,CG",
            "A139": "OG,CB,CA",
            "A152": "OH,CZ",
            "A156": "NZ,CE,CD",
            "ACT": "OXT",
            "NAI": ""
        }
    }
}
```

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Hugging Face Spaces

Low-code: Interacting with Spaces through the gradio API ensures you do not accidentally lose the job, and can help organize outputs with Python scripting.

example scripts of interacting with the API coming soon to the github repo!

API documentation hugging-science/RFdiffusion3 API Recorder 7 API endpoints

Choose one of the following ways to interact with the API.

Copy Page

Python JavaScript cURL MCP

1. Install the Python client (docs) if you don't already have it installed.

```
$ pip install gradio_client
```

2. Find the API endpoint below corresponding to your desired function in the app. Copy the code snippet, replacing the placeholder values with your own input data. If this is a private Space, you may need to pass your Hugging Face token as well (read more). Or use the API Recorder to automatically generate your API requests.

API name: /give_run_status Once the user presses on the Run Generation button, this function checks whether both config and scaffold are ready, and gives a status message about the generation run that is about to happen. If both config and scaffold are not ready, it prompts the user to ensure they are both validated before running the generation.

12 requests (67% successful, p50: 0 ms)

```
from gradio_client import Client, handle_file

client = Client("hugging-science/RFdiffusion3")
result = client.predict(
  num_batches=5,
  num_designs_per_batch=2,
  config_upload=handle_file('https://github.com/gradio-app/gradio/raw/main/test/test_files/sample_f:
  scaffold_upload=handle_file('https://github.com/gradio-app/gradio/raw/main/test/test_files/sample_
  api_name="/give_run_status"
)
print(result)
```

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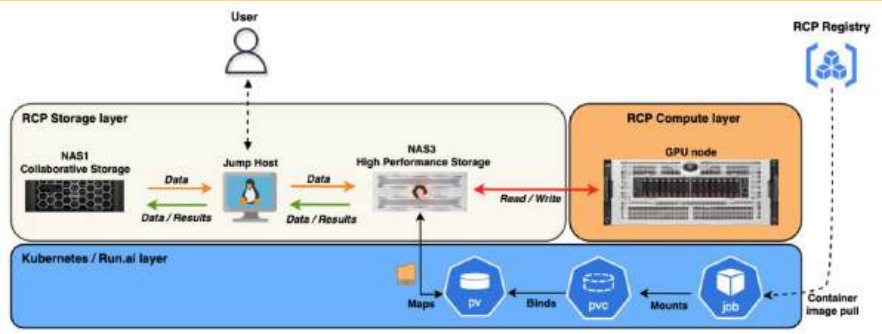
Research computing platform

***Advantages**:*

- Big jobs, can even run overnight!
- Can develop code on the cluster to directly process the outputs of models.
- Try to install or fine-tune new models if you want.

***Disadvantages**:*

- Time to install all necessary tools (docker, kubernetes, run:ai)
- Overhead to get used to syntax of submitting jobs
- Need to be an EPFL student connected to the EPFL network (or VPN)



More documentation and tutorials on how to run RCP **coming on our Github repository.**

Workshop Saturday morning on using RCP

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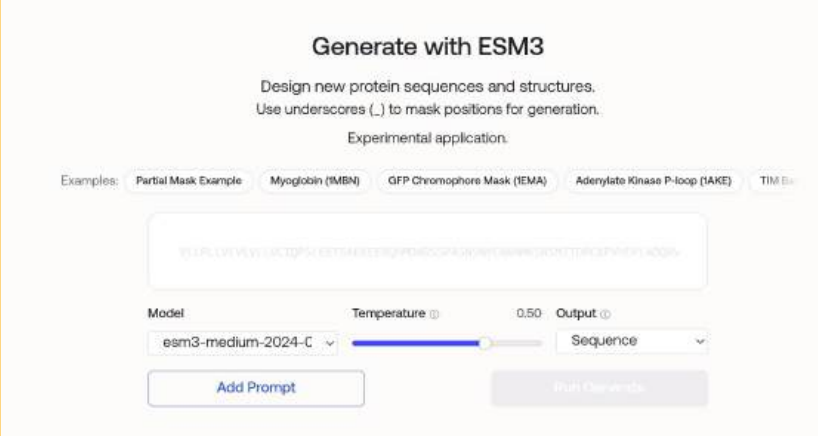
Online tools

Advantages:

- Easy-to-use
- Running on the company's server

Disadvantages:

- Rate /job submission limits
- Access to limited GPUs (T4 on Google Colab)
- Need to create an account



https://platform.latentlabs.com/

https://www.evolutionaryscale.ai/

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Resume of tech infrastructure

Hugging Face Spaces 🤗

- Quick prototyping for 10's or a couple 100's of designs.
- refine your input configurations, quickly try different binding interfaces
- main tool for students outside EPFL.

RCP🤖

- heavy jobs with 100's to 1000's of design candidates
- create pipelines and post-process outputs all on the cluster

Your laptop💻

- run multiple HF Spaces jobs with Gradio API
- post-process outputs of HF spaces
- submit RCP jobs from your terminal
- literature, PDB and Uniprot research
- visualize target and most promising designs **with Pymol**

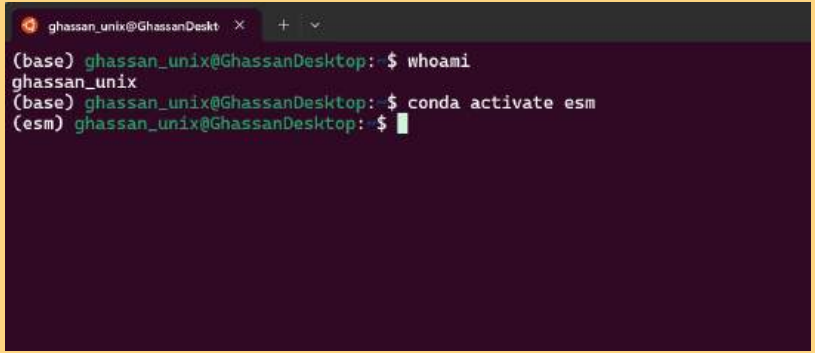
Our Git summarizes the different tools available for you !

PDW

A note on using your laptop

- If you're on Mac, good. If you're on Linux, perfect. **If you're on Windows, install WSL2** → Linux system inside Windows without dual-boot

- Careful with **disk space!** a WSL installation is heavy and disk space is not reclaimed after file deletion.



WSL2: 90 Gb, rest: 100 Gb

↓ download 5 GB file

WSL2: 95 Gb, rest: 100 Gb

↓ delete 5 GB file

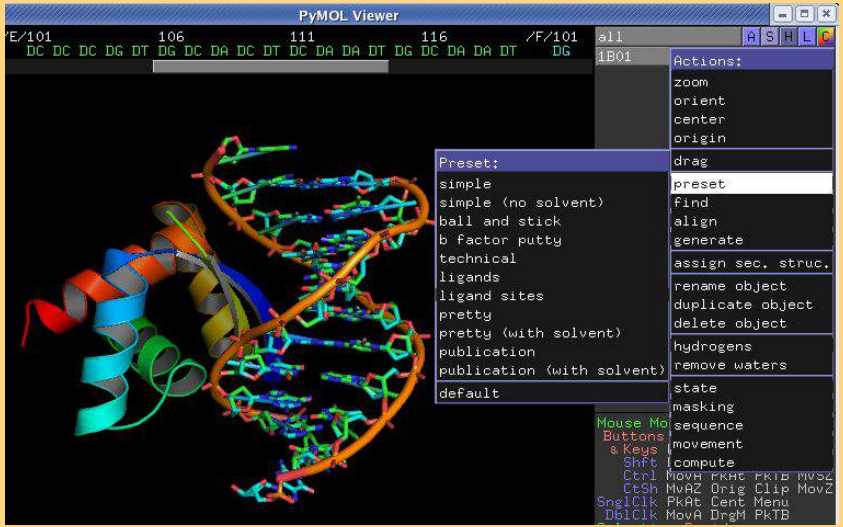
WSL2: 95 Gb, rest: 100 Gb

PDW

Pymol

- **Essential tool for any protein design project:** visualization, selection, cropping, measuring, aligning

- Tutorial by Emma coming soon to the Github repo + LLMs are great at it



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Advice on team workflow

- Teams are evenly balanced between CS and biology specialists, and between years of study.

- Some parts of the pipeline have considerable overhead
  - setting up first RFD3 config
  - connecting to RCP

Divide-and-conquer approach

- some focus on biological question, finding and visualizing target structure, setting up design config
- some focus on getting RCP working for the team
- some experiment on Spaces

BUT be mindful of your teammates' preferences. This is more a learning experience than a competition!

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TO DO list

Before the hackathon...

- ✅ **Install PyMol** (or other protein visualization tool): https://pymol.org/
- ✅ **Install Docker, Kubectl and Run:ai** (EPFL students only): tutorial here (Part 1)
- ✅ **Create a HuggingFace account:** https://huggingface.co,
  Submit your Hugging Face username here for expanded rate limits
  Join the Hugging Science organization: HuggingScience
- ✅ Fill this form so we can use Alphafold3 without getting sued by Google Deepmind. (**mandatory**)
- ✅ Read your associated **survival guide on Github.** (optional)
  Come to **office hours** tomorrow if you have any questions (very optional).

PDW

Good Luck and Enjoy the Ride!

PDW